

# Anass Badreddine El Hallaoui

Citizen

M.Sc. Candidate (Graduating Spring 2028) — B.Sc. Completed Jun 2026

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## EXPERIENCE

### Shopify

Jan 25 – Jul 25

Software Engineer Intern (Visual Search)

May 25 – Jul 25

- **Co-created** Shopify's **Visual Search Division** during an intern hackathon, leveraging rapid prototyping and experimentation to win **1st place** and secure **executive sponsorship**.
- **Championed** the cross-platform delivery of Visual Search by driving a **10-person squad**, taking proactive ownership to launch to **25M+ users** in **6 weeks** and presenting to the **CEO**.
- Architected a production-ready **Computer Vision similarity pipeline**, optimizing inference to process **55M+ images** with **<3s latency** while operating at **0.1% of competitors' compute costs**.
- Established scalable **observability infrastructure** across **10 Shop App products** (including Visual Search), implementing **30+ metrics**, **9 dashboards**, and **50+ proactive alerts** with Grafana & GCP to ensure enterprise-wide reliability.

Software Engineer Intern (Infrastructure Security Team)

Jan 25 – Apr 25

- **Engineered** a **gRPC-based proof of concept**, achieving **10×** faster internal requests and **90%** lower infrastructure costs.
- Contributed to **Redpanda's open-source codebase** by implementing **SAN URI support**, enabling Shopify's integration and earning **recognition from Redpanda's CEO and Shopify's VPs**.
- **Resolved** a **critical GCP firewall configuration** issue, restoring secure connectivity to **75+ cloud resources** across the organization.

### Intel

Feb 24 – Sep 25

oneAPI Software Engineer Ambassador

- Directed technical workshops on Intel AI tooling across 10+ universities and hackathons, upskilling **1,000+ students**.
- Engineered UI/UX enhancements for the Intel Tiber Cloud interface, improving overall platform performance by **15%**.

### iA Financial Group

May 23 – Sep 23

Software Engineer Intern

- Modernized a legacy payroll system, accelerating processing speed by **87%** for 17,000+ weekly salaries totaling **\$20M+**.
- Identified and patched a critical privacy vulnerability, safeguarding sensitive financial data for **10,000+ clients** and employees.
- Engineered a robust insurance-management feature, reducing operational errors by **50%** and boosting workflow efficiency by **35%**.

### OpenAI

Nov 21 – Jul 22

AI Research

- Evaluated **early GPT-3.5 foundational models** for alignment and edge-case behavior, providing structured, high-fidelity insights for top researchers.

## LEADERSHIP & AWARDS

- Won **NASA, STANFORD, SHOPIFY** + 5 other **hackathons**.
- **GitHub Community Wall of Famer 2025**.
- Recipient of **\$80K+** in scholarships, including a full-ride U.S. exchange semester.
- Member of the 2024 **Putnam International Math Competition** team.

## PROJECTS

### AgentGym (AARL) – Self-Improving Agentic AI Framework

GitHub

Open-Source Framework

- Engineered an **AI-native** self-improving framework, implementing **Hindsight Experience Replay (AgentHER)** and DPO to **increase autonomous task success by 38%** while **cutting compute costs by 98%**.
- Developed a transparent LLM proxy to **validate AI-generated outputs** and investigate failures, capturing **10,000+ trajectories** to refine agent behaviors using Mantel test correlations.

### NeuroVision – AI Video Optimization via Human Brain-Response Prediction

GitHub

Open-Source Project

- Developed an **AI video analyzer** that predicts how an audience's brain will respond to content, providing automated suggestions to optimize **visual engagement**.
- Engineered a **multimodal PyTorch pipeline** (Meta TRIBE v2) that processes video/audio into **20,000+ cortical data points** per second using a **48GB VRAM** inference engine.

TECHNICAL SKILLS

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**Programming & Scripting:** Java, Python, C++, C#, Rust, Ruby, Go, SQL, TypeScript, JavaScript, Swift, Kotlin  
**Frontend & Mobile:** React, React Native, Electron.js, Bootstrap, HTML/CSS, Yarn, SwiftUI, Xcode, Android SDK/Studio, Jetpack Compose, Dagger, Flutter  
**Backend, APIs & Cloud:** Node.js, Flask, Ruby on Rails, ASP.NET/.NET, GraphQL, gRPC, REST APIs, Kubernetes, AWS, Azure, GCP, Docker, Terraform, Kafka, Redpanda, Grafana, Prometheus  
**Databases & Data:** MySQL, Postgres, MongoDB, NoSQL, Redis, BigQuery, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Apache Spark, CUDA, Data Structures & Algorithms  
**Practices & Tools:** OOP, SOLID, TDD, Unit Testing, Design Patterns, IaC, Agile, CI/CD (Buildkite, Bitrise, GitHub Actions), Git, ServiceNow

EDUCATION

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|------------------------------------------------------|--------|
| <b>Georgia Institute of Technology</b>               |        |
| M.Sc. Computer Science (AI Specialization)           |        |
| <b>Laval University</b>                              | Jun 26 |
| B.Sc. Computer Science (GPA: 3.7)                    |        |
| <b>The Ohio State University - Exchange Student</b>  | May 24 |
| B.Sc. Computer Science (Data Science Specialization) |        |